

CSE392

Seminar

Ethical Principles in IT

Presentation Content

- Ethics and Engineering Ethics
 - Ethics and Morality
 - Professional Ethics
 - Engineering Ethics
- Ethical Issues in Computer Engineering
- IT Ethics and Law

Ethics and Engineering Ethics

What is Ethics?

- At its simplest, ethics is a system of moral principles.
 - They affect how people make decisions and lead their lives.
- Ethics is concerned with what is good for individuals and society and is also described as moral philosophy.
- The term is derived from the Greek word *ethos* which can mean custom, habit, character or disposition.

What is Ethics? (Cont.)

- Ethics covers the following dilemmas:
 - how to live a good life
 - our rights and responsibilities
 - the language of right and wrong
 - moral decisions - what is good and bad?
- Our concepts of ethics have been derived from religions, philosophies and cultures. They infuse debates on topics like abortion, human rights and professional conduct.

Ethics vs. Morality - What're the differences?

- Morality is a person or society's idea of what is right or wrong, especially in regard to a person's behavior.
- Many scholars and researchers don't differentiate between morals and ethics, and that's because they're very similar. Many definitions even explain ethics as a set of moral principles.
- In general, morals are considered guidelines that affect individuals, and ethics are considered guideposts for entire larger groups or communities. Ethics are also more culturally based than morals.

Professional Ethics

- Professional ethics are principles that govern the behaviour of a person or group in a business environment.
- In other words, professional ethics are standards set by professional organizations for the behavior and values of people working within a specific field.
- Codes of professional ethics are established in order to provide guidance to professionals, usually to not abuse client relationships and preserve the integrity and reputation of the applicable profession.

Professional Ethics (Cont.)

- When a person violates one or more of these promulgated ethics, he or she is subject to disciplinary action by the governing body of the profession.
- A high level of adherence to a code of professional ethics is needed so that the public will be confident in the moral standards of those working in the field. Codes of professional ethics are commonly applied to professionals in the fields of accounting, law, and medicine.

Professional Ethics vs. Values - What're the similarities and differences?

- Like values, professional ethics provide rules on how a person should act towards other people and institutions in such an environment.
- Unlike values, professional ethics are often codified as a set of rules, which a particular group of people use.
- This means that all those in a particular group will use the same professional ethics, even though their values may be unique to each person.

What is engineering ethics?

- (1) the study of moral issues and decisions confronting individuals and organizations involved in engineering.
- (2) the study of related questions about moral conduct, character and relationships of peoples and organizations involved in technological development.

What are the purposes for Code of Ethics for Engineers?

- Provide positive stimulus for ethical conduct
- Helpful advice concerning the primary obligation of “engineers”
- Guideposts in interpreting ethical dilemmas

Code of Ethics that Apply to Computer Engineers?

- IEEE Code of Ethics
- IEEE-CS/ACM Software Engineering Code of Ethics and Professional Practice
- National Society of Professional Engineers Code of Ethics

Preamble NSPE Code of Ethics

- Engineering is an important and learned profession.
- Engineers are expected to exhibit the highest standards of honesty and integrity.
- The services provided by engineers require honesty, fairness and equity, and must be dedicated to the protection of the public health, safety, and welfare.
- Engineers must perform under a standard of professional behavior that requires adherence to the highest principles of ethical conduct.

General Rules

1. Hold paramount the safety and welfare of the public.
2. Perform services only in areas of their competence.
3. Issue public statements only in an objective and truthful manner.
4. Act for each employer or client as faithful agents or trustees.
5. Avoid deceptive acts.
6. Conduct themselves ethically, and lawfully so as to enhance the honor and usefulness of the profession.

Rules of Practice

1. Engineers shall hold paramount the safety, health, and welfare of the public.
 - a. If engineers' judgment is overruled under circumstances that endanger life or property, they shall notify their employer or client and such other authority as may be appropriate.
 - b. Engineers shall approve only those engineering documents that are in conformity with applicable standards.
 - c. Engineers shall not reveal facts, data or information without the prior consent of the client or employer except as authorized or required by law or this Code.
 - d. Engineers shall not permit the use of their name or associate in business ventures with any person or firm that they believe are engaged in dishonest enterprise.
 - e. Engineers shall not aid or abet the unlawful practice of engineering by a person or firm.
 - f. Engineers having knowledge of any alleged violation of this Code shall report thereon to appropriate professional bodies and cooperate with the proper authorities in furnishing such information or assistance as may be required.

Rules of Practice

2. Engineers shall perform services only in the areas of their competence.

a. Engineers shall undertake assignments only when qualified by education or experience in the specific technical fields involved.

b. Engineers shall not affix their signatures to any plans or documents dealing with subject matter in which they lack competence, nor to any plan or document not prepared under their direction and control.

c. Engineers may accept assignments and assume responsibility for coordination of an entire project and sign and seal the engineering documents for the entire project, provided that each technical segment is signed and sealed only by the qualified engineers who prepared the segment.

Rules of Practice

3. Engineers shall issue public statements only in an objective and truthful manner.

a. Engineers shall be objective and truthful in professional reports, statements, or testimony. They shall include all relevant and pertinent information in such reports, statements, or testimony, which should bear the date indicating when it was current.

b. Engineers may express publicly technical opinions that are founded upon knowledge of the facts and competence in the subject matter.

c. Engineers shall issue no statements, criticisms, or arguments on technical matters that are inspired or paid for by interested parties, unless they have prefaced their comments by explicitly identifying the interested parties on whose behalf they are speaking, and by revealing the existence of any interest the engineers may have in the matters.

Rules of Practice

4. Engineers shall act for each employer or client as faithful agents or trustees.

a. Engineers shall disclose all known or potential conflicts of interest that could influence or appear to influence their judgment or the quality of their services.

b. Engineers shall not accept compensation, financial or otherwise, from more than one party for services on the same project, unless the circumstances are fully disclosed and agreed to by all interested parties.

c. Engineers shall not solicit or accept financial or other valuable consideration, directly or indirectly, from outside agents in connection with the work for which they are responsible.

d. Engineers in public service as members or employees of a governmental body or department shall not participate in decisions with respect to services solicited or provided by them or their organizations in private or public engineering practice.

e. Engineers shall not accept a contract from a governmental body on which a principal or officer of their organization serves as a member.

Rules of Practice

5. Engineers shall avoid deceptive acts.

a. Engineers shall not falsify their qualifications or permit misrepresentation of their or their associates' qualifications. They shall not misrepresent or exaggerate their responsibility in or for the subject matter of prior assignments. Brochures or other presentations incident to the solicitation of employment shall not misrepresent pertinent facts concerning employers, employees, or past accomplishments.

b. Engineers shall not offer, solicit or receive, either directly or indirectly, any contribution to influence the award of a contract by public authority, or which may be reasonably construed by the public as having the effect of intent to influencing the awarding of a contract. They shall not offer any gift or other valuable consideration in order to secure work. They shall not pay a commission, percentage, or brokerage fee in order to secure work.

Professional Obligations

1. Engineers shall be guided in all their relations by the highest standards of honesty and integrity.
2. Engineers shall at all times strive to serve the public interest.
3. Engineers shall avoid all conduct or practice that deceives the public.
4. Engineers shall not disclose, without consent, confidential information concerning the business affairs or technical processes of any present or former employer on which they serve.
5. Engineers shall not be influenced in their professional duties by conflicting interests.

Professional Obligations

6. Engineers shall not attempt to obtain employment or advancement by untruthfully criticizing other engineers.
7. Engineers shall not attempt to injure or falsely the professional reputation of other engineers. Engineers who believe others are guilty of unethical shall present such information to the proper authority for action.
8. Engineers shall accept personal responsibility for their professional activities, provided, however, that Engineers may seek indemnification for services arising out of their practice for other than gross negligence, where the Engineer's interests cannot otherwise be protected.
9. Engineers shall give credit for engineering work to those to whom credit is due, and will recognize the proprietary interests of others.

Ethical Issues in Computer Engineering

There are many ethical issues in computer engineering field. To deal with these issues in information technology society, it is important to find out the ethical issues. Some of the major ethical issues faced by computer engineering are:

- Personal Privacy
- Access Right
- Harmful Actions
- Patents
- Copyright
- Liability
- Piracy

Personal Privacy

- Once data is put into a computer it can easily be copied or transmitted.
- This puts people's personal data at risk and organisations that hold such data should do their best to ensure it remains private.
- Data protection legislation exists to ensure that organisations in the UK comply with the law regarding data privacy.

Access Right

- Access right becomes a high priority issue for the IT and cyberspace with the great advancement in technology.
- E-commerce and Electronic payment systems evolution on the internet heightened this issue for various corporate organizations and government agencies.
- Network on the internet cannot be made secure from unauthorized access. Generally,
- the intrusion detection system are used to determine whether the user is an intruder or an appropriate user.

Harmful Actions

- Harmful actions in the computer ethics refers to the damage or negative consequences to the IT such as loss of important information, loss of property, loss of ownership, destruction of property and undesirable substantial impacts.
- This principle of ethical conduct restricts any outsiders from the use of information technology in manner which leads to any loss to any of the users, employees, employers and the general public.
- Typically, these actions comprises of the intentional destruction or alteration of files and program which drives a serious loss of resources. To recover from the harmful actions extra time and efforts are required to remove the viruses from the computer systems.

Patents

- A patent can preserve the unique and secret aspect of an idea.
- Obtaining a patent is very difficult as compared with obtaining a copyright.
- A thorough disclosure is required with the software.
- The patent holder has to reveal the full details of a program to a proficient programmer for building a program.

Copyright

- Copyright law works as a very powerful legal tool in protecting computer software, both before a security breach and surely after a security breach.
- Breach of the copyright could be the mishandling and misuse of data, computer programs, documentation and similar material.
- In many countries, copyright legislation is amended or revised to provide explicit laws to protect computer programs.

Liability

- One should be aware of the liability issue in making ethical decisions.
- Software developer makes promises and assertions to the user about the nature and quality of the product that can be restricted as an express warranty.
- Programmers or retailers possess the legitimate to determine the express warranties. Thus they have to be practical when they define any claims and predictions about the capacities, quality and nature of their software or hardware.
- Every word they say about their product may be as legally valid as stated in written. All agreements should be in writing to protect against liability.

Piracy

- Piracy is an activity in which the creation of illegal copy of the software is made.
- It is entirely up to the owner of the software as to whether or not users can make backup copies of their software.
- As laws made for copyright protection are evolving, also legislation that would stop unauthorized duplication of software is in consideration.

Information Technology (IT) Ethics and Law

Information technology (IT)

- Information technology (IT) is the use of computers to create, process, store, retrieve and exchange all kinds of electronic data and information
- Included in Information Technology
 - All computers with a human interface
 - All computer peripherals which will not operate unless connected to a computer or network
 - All voice, video and data networks and the equipment, staff and purchased services necessary to operate them
 - All salary and benefits for staff whose job descriptions specifically includes technology functions, i.e. network services, applications development, systems administration
 - All technology services provided by vendors or contractors
 - Operating costs associated with providing information technology
 - All costs associated with developing, purchasing, licensing or maintaining software

Information Technology (IT) Ethics

- Information technology ethics is the study of the ethical issues arising out of the use and development of electronic technologies.
- 5 Legal and Ethical Issues in IT
 - Privacy
 - Digital Ownership
 - Data Gathering
 - Security Liability
 - Access Costs

Privacy

- Most people have their personal data spread throughout the digital world.
- Even things thought to be secure, such as email or private accounts, can be accessed by unintended sources
- Privacy has evolving legal implications, but there are also ethical considerations
- Do people know how their accounts are monitored? To what extent is such monitoring occurring?
- Privacy concerns can easily become a slippery slope, slowly eroding an individual's right to privacy completely.

Digital Ownership

- Digital mediums have allowed information to flow more freely than before. This exchange of ideas comes with a legal and ethical backlash.
- How can ownership be established in the digital realm? Things can be easily copied and pasted online, which makes intellectual property hard to control.
- Legal notions such as copyright have struggled to keep up with the digital era.
- Companies in the music and entertainment industries have pushed for greater legal protections for intellectual properties while other activists have sought to provide greater freedoms for the exchange of ideas in the digital realm

Data Gathering

- On some level, everyone knows that their online lives are monitored. The United States has even passed legislation allowing the government to actively monitor private citizens in the name of national security.
- These measures have revived a debate about what information can be gathered and why.
- This debate applies on a smaller scale as well because companies need to consider what information to collect from their employees
- This issue invokes a question of consent. Do people know what information is being monitored? Do they have a right to know how their data is being used?

Security Liability

- In the past, security issues were resolved by locking a door. Digital security is much more complicated.
- Security systems for digital networks are computerized in order to protect vital information and important assets. However, this increased security comes with increased surveillance.
- All security systems have inherent risks, which means it is a question of what risks are acceptable and what freedoms can be forfeited.
- Ultimately, IT professionals need to balance risk with freedom to create a security system that is effective and ethical at the same time.

Access Costs

- Net neutrality has become a trendy issue thanks to legislative efforts in the last few years. The issue of net neutrality is essentially a question of access.
- Proponents want the Internet to remain open to everyone while some businesses want to create tiered access for those who are willing to pay.
- The issue even extends to private Internet usage since the cost of service in some areas may be cost prohibitive.
- The larger ethical question is whether or not digital exchange is now a universal right. The cost of access can impede business growth, entrepreneurial spirit and individual expression.

Ethical standards in computer

Various national and international professional societies and organizations have produced code of ethics documents to give basic behavioral guidelines to computing professionals and users. They include:

- Association for Computing Machinery
 - ACM Code of Ethics and Professional Conduct
- Australian Computer Society
 - ACS Code of Ethics
 - ACS Code of Professional Conduct
- British Computer Society
 - BCS Code of Conduct
 - Code of Good Practice
- Computer Ethics Institute
 - Ten Commandments of Computer Ethics
- IEEE
 - IEEE Code of Ethics
 - IEEE Code of Conduct
- League of Professional System Administrators
 - The System Administrators' Code of Ethics

Ten Commandments of Computer Ethics

The Ten Commandments of Computer Ethics were created in 1992 by the Washington, D.C. based Computer Ethics Institute

1. Do not use the computer in ways that may harm other people.
2. Do not use computer technology to cause interference in other users' work.
3. Do not spy on another person's computer data.
4. Do not use computer technology to steal information.
5. Do not contribute to the spread of misinformation using computer technology.
6. Refrain from copying software or buying pirated copies. Pay for software unless it is free.
7. Do not use someone else's computer resources unless authorized to.
8. It is wrong to claim ownership on a work which is the output of someone else's intellect.
9. Before developing a software, think about the social impact it can have.
10. In using computers for communication, be respectful and courteous with the fellow members.

Computer Crime

- Alternatively referred to as cyber crime, e-crime, electronic crime, or hi-tech crime.
- Computer crime is an act performed by a knowledgeable computer user, sometimes referred to as a hacker that illegally browses or steals a company's or individual's private information.
- In some cases, this person or group of individuals may be malicious and destroy or otherwise corrupt the computer or data files.

Computer Crime (Cont.)

- The growth and advances in digital technology creates a whole new platform for criminal activity.
- Since the advancement of technology, any crime that involves using a computer or network is generally referred to as a cybercrime or computer crime.
- The penalties differ for each crime, depending on whether they violated state or federal laws. In general, they include fines, imprisonment, probation, or all the above.

Common Types of Computer Crimes

Below is a list of the different types of computer crimes today.

- Child Pornography/Abuse
- Identity Theft
- Cyberbully or Cyberstalking
- Hacking
- Creating Malware
- Espionage
- Phishing
- Spoofing

Child Pornography/Abuse

- This cybercrime can involve the perpetrator looking to create or distribute sexual images of children.
- In some cases, the accused seeks out minors on the internet, whether that be via social media or chatrooms with the objective of producing child pornography.

Identity Theft

- Identity theft is the act of a person obtaining information illegally about someone else.
- Thieves try to find such information as full name, maiden name, address, date of birth, social security number, passwords, phone number, e-mail, and credit card numbers.
- The thief can then use this information to gain access to bank accounts, e-mail, smartphones, identify themselves as you, or sells your information.

Cyberbully or Cyberstalking

- Alternatively referred to as a cyberstalker, a cyberbully is someone who posts inappropriate or unwanted things about another person, or otherwise harasses them in e-mails, IMs, or SMS.
- A cyberbully may try to ruin another person's reputation by posting false information, or they may post information the other person does not want others to know.
- Online social media platforms, like Facebook and Twitter, are common places for a cyberbully to harass other people.

Hacking

- The computer crime hacking refers to the practice of gaining unauthorized access to another person's computer, database, or network where private information can be stored.
- It is a felony to hack a computer system, whether it is a single personal computer or an organizational computer network.
- However, not all types of “hacking” refer to crimes.
- Some organizations perform “Ethical hacking” on their own systems or with permission to explore the systems of others to look for vulnerabilities. This is treated differently than “malicious hacking,” the act of entering someone's computer without their knowledge to take data or leave viruses.

Creating Malware

- Malware is malicious software developed to change your settings, delete software, cause errors, watch browsing habits, or open a computer to attacks. Malware can also cause harm by potentially disabling or disrupting computers, which could affect humans' health and welfare. It uses deceptive and unethical tactics to install itself on your computer without your consent. For instance, you may visit a website and receive an unrequested download, and mistakenly run this software on your computer. It may also use vulnerabilities in your operating system or web browser to install itself without requiring you to manually run the software.
- Malware is commonly found on websites that engage in illegal or questionably legal activities, such as downloading copyrighted videos, music, or software for free. Many of these sites may offer services for free, drawing people in to use their services or download their applications. Those services or applications are often infected with malware, which then infects your computer.

Espionage

- Espionage or spying is the act of obtaining secret or confidential information from non-disclosed sources or divulging of the same without the permission of the holder of the information. A person who commits espionage is called an espionage agent or spy.
- Any individual or spy ring (a cooperating group of spies), in the service of a government, company, criminal organization, or independent operation, can commit espionage.

Phishing

- Pronounced like fishing, phishing is a term used to describe a malicious individual or group who scam users.
- They do so by sending e-mails or creating web pages designed to collect an individual's online bank, credit card, or other login information.
- Because these e-mails and web pages look legitimate users trust them and enter their personal information.

Spoofing

- In general, the term spoof refers to hacking or deception that imitates another person, software program, hardware device, or computer to bypass security measures.
- The most commonly known types of spoofing:
 - IP spoofing
 - E-mail or address spoofing
 - Phone number or caller ID spoofing
 - Web pages spoof

Information Technology (IT) Law

- The law is the principles and regulations established in a community by some authority and applicable to its people, whether in the form of legislation or of custom and policies recognized and enforced by judicial decision.
- Information technology law (also called cyberlaw) concerns the law of information technology, including computing and the internet.
- It is related to legal informatics, and governs the digital dissemination of both (digitized) information and software, information security and electronic commerce aspects and it has been described as "paper laws" for a "paperless environment".
- It raises specific issues of intellectual property in computing and online, contract law, privacy, freedom of expression, and jurisdiction.

Information Technology (IT) Law (Cont.)

- IT law does not constitute a separate area of law rather it encompasses aspects of contract, intellectual property, privacy and data protection laws. Intellectual property is an important component of IT law, including copyright, rules on fair use, and special rules on copy protection for digital media, and circumvention of such schemes.
- The related topics of software licenses, end user license agreements, free software licenses and open-source licenses can involve discussion of product liability, professional liability of individual developers, warranties, contract law, trade secrets and intellectual property.

Information Technology (IT) Law (Cont.)

- In other words, the term IT law refers to an ever-growing field of law that focuses on those legal issues that arise from the emerging information society, which is driven by certain technologies like computers, wireless communications and the Internet.
- Topics encompassed within information technology law include:
 - Protection of computer software and databases
 - Information access and controls
 - Privacy and security
 - Internet law
 - Electronic commerce

Information Technology (IT) Law (Cont.)

- Information technologies are extremely powerful tools.
- Therefore, the stakes of identifying the best laws and policies for their use are very high.
- These stakes fall into three general categories:
 - 1) Economic
 - 2) Social and Personal
 - 3) Political

Economic

- Information technologies and information-based products and services are becoming central to the economy as a whole.
- The new technologies and the information they embody can be used to improve efficiency, increase productivity, and thus engender economic growth.
- Information is reusable, and unlike capital resources, such as steel or iron, it can be produced and distributed using few physical resources.
- Not only is information an efficient substitute for labor, it can also be used to improve the overall efficiency of the production process.
- Businesses, for example, are now applying information technology to almost all of their activities: from recruiting to laying off workers, from ordering raw materials to manufacturing products, from analyzing markets to performing strategic planning, and from inventing new technologies to designing applications for their use. To serve these needs, whole new industries have been spawned.

Economic (Cont.)

- One of the fastest growing sectors of the economy, the information industry is spearheading national and international economic growth and enhancing every country's competitive position in the international marketplace.
- The economic stakes raised by the new technologies are particularly high for the copyright industries — publishing and other industries that rely on the legal protections provided by copyright law.
- The amount of financial damage that these industries suffer due to infringements of intellectual property rights is extremely hard to estimate.

Social and Personal

- Information is a dominant force in our lives. In the United States, an enormous amount of information is communicated in the form of words through electronic media. In the 1970s, for example, it was estimated that the American population was exposed to about 8.7 trillion words each day through electronic media such as radio, television, and print media such as newspapers, books, and magazines.
- This figure has risen each year; the number of words communicated increases at an average rate of 1.2% per year. And the fastest growing media for transmitting and receiving information today is the Internet.

Political

- In democratic societies, citizens must be well informed about issues, candidates for office, and local affairs. Similarly, a democratic polity requires a well-informed citizenry. Increasingly, information and communications technologies serve these information needs. The government regularly needs huge amounts of information to make complex legal and policy decisions. Many government agencies would find it impossible to conduct their daily business without resorting to customized information on demand.
- The Internal Revenue Service and the Social Security Administration, for example, require large automated information systems to handle the accounts of hundreds of millions of clients. And the operation of national defense depends on the use of complex communications systems both for day-to-day management of the military establishment and for the command and control of sophisticated weaponry.

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